

Phase1_Daily_Schedule

Phase 1 — Day-by-Day Schedule · Mathematics + NNs from First Principles

Weeks 3-12 · Mon Sep 21 → Sun Nov 29, 2026 · ~210 hrs · *the bedrock — do NOT rush*

Goal: real mathematical maturity — hand-derive backprop, implement micrograd from memory, whiteboard attention with the $\sqrt{d_k}$ rationale. The Phase-1 self-test (Week 10) is the gate that switches the Threads on.

Blocks: **A** 06:00-08:00 (hardest new theory, derive by hand) · **B** 08:30-10:30 (build in code) · **Evening** 20:30-22:30 (re-watch, Anki, glossary, plan tomorrow). **Weekend** = ~10 h buffer (spillover/revision/rest — protect one full day off). **Threads (start Week 10, inside the existing blocks):** **T** = DSA+C++ (first ~45 min of Block B) · **M** = implement-from-scratch (Block B) · **R** = reproduce-a-paper + write + apply (Evening, ~2 h/wk). (*Phase 0 is already covered in the Quarter-1 schedule.*)

Week 3 · Sep 21-27 — Linear Algebra I

Day	Block A (06-08)	Block B (08:30-10:30)	Evening (20:30-22:30)
Mon	3B1B LA ch 1-2: vectors, span, linear combinations	NumPy: vectors, linear combos, visualize	Re-watch 3B1B; Anki (LA terms); glossary
Tue	3B1B: linear transformations, matmul as composition	Implement matmul + 2D transforms; plot	Anki; glossary
Wed	Determinants (3B1B) + MIT 18.06 L1-3	Implement determinant; test invertibility	Anki; glossary
Thu	Inverse, rank, column/null space (18.06 L4-6)	Solve $Ax=b$; rank experiments	Anki; glossary
Fri	Dot product, projections (18.06 L14-15)	Implement projection onto a subspace	Whiteboard-Fri: projection formula; teach-a-junior note
Sat-Sun	Buffer — finish spillover · light review · 1 full rest day		

Week 4 · Sep 28-Oct 4 — Linear Algebra II (eigen / SVD — the crux)

Day	Block A	Block B	Evening
Mon	Eigenvalues/eigenvectors (3B1B + 18.06 L21)	Implement eigendecomposition; verify	Anki (eigen); glossary
Tue	Change of basis, diagonalization (L22)	Diagonalize; power iteration	Anki
Wed	SVD (18.065) — derive from eigendecomp of $A^T A$	Implement SVD; compare to numpy	Anki (SVD — triangulate if slippery)
Thu	Rank-r approx (Eckart-Young); PSD	Truncated-SVD image compression	Anki
Fri	SVD applications; PCA preview	SVD-based PCA on a toy set	Whiteboard-Fri: SVD-from-eigendecomp; why $A^T A$ is PSD
Sat-Sun	Buffer + rest		

Week 5 · Oct 5-11 — 🌀 CONSOLIDATION (no new material, ~18 h)

Day	Block A	Block B	Evening
Mon-Fri	Re-derive from memory: matmul-as-composition, determinant, SVD-from-eigendecomp, projection	Re-implement (blank file): SVD-PCA + a subspace projection	Full Anki + glossary catch-up; Fri: teach-a-junior note on SVD
Sat-Sun	Rest + optional buffer for any LA gap		

Week 6 · Oct 12-18 — LA finish + Calculus / matrix-calculus I

Day	Block A	Block B	Evening
Mon	PSD, quadratic forms (18.06)	Implement quadratic-form eval; check PD	Anki
Tue	3B1B Calculus: derivatives, chain rule	Numeric derivative + finite-diff check	Anki
Wed	Partial derivatives, gradients	Gradient of a scalar field; plot	Anki
Thu	Matrix calculus (Parr & Howard): $\partial(Wx)$	Implement $\partial L/\partial W = \delta x^T$; grad-check	Anki
Fri	Jacobians	Jacobian of a vector function	Whiteboard-Fri: backprop through a linear layer
Sat-Sun	Buffer + rest		

Week 7 · Oct 19-25 — Matrix-calc II + Probability I

Day	Block A	Block B	Evening
Mon	Softmax + cross-entropy gradient (derive p−y)	Implement softmax-CE fwd/bwd; grad-check	Anki
Tue	Stat 110: axioms, conditional probability	Simulate conditional probability	Anki
Wed	Bayes' theorem	Bayes worked example in code	Anki
Thu	Random variables, expectation	Simulate E[X], Var[X]	Anki
Fri	Variance, covariance	Covariance matrix in code	Whiteboard-Fri: Bayes + softmax-CE gradient
Sat-Sun	Buffer + rest		

Week 8 · Oct 26-Nov 1 — Probability II

Day	Block A	Block B	Evening
Mon	Distributions (Bernoulli/Binomial/Gaussian)	Sample & fit distributions	Anki
Tue	MLE — derive Gaussian mean/variance	Implement an MLE fit	Anki
Wed	Conditional expectation as projection	E[X	Y] demo in code
Thu	CLT (proof sketch)	CLT simulation	Anki
Fri	Entropy / cross-entropy / KL	Implement entropy, CE, KL	Whiteboard-Fri: E[X
Sat-Sun	Buffer + rest		

Week 9 · Nov 2-8 — Probability finish + NN-from-scratch I (micrograd)

Day	Block A	Block B	Evening
Mon	Finish key probability (covariance, moments)	Wrap probability drills	Anki
Tue	Karpathy Z2H: micrograd (autodiff intuition)	Start micrograd (the Value class)	Re-watch micrograd
Wed	Backprop as chain rule over a graph	micrograd: the backward pass	Anki (autodiff)
Thu	micrograd: a tiny MLP	micrograd: build + train an MLP	Anki
Fri	Review autodiff	Finish + test micrograd MLP	Whiteboard-Fri: backprop through an MLP; note
Sat-Sun	Buffer + rest		

Week 10 · Nov 9-15 — NN II + THREADS BEGIN + Phase-1 self-test

Day	Block A	Block B (T/M + build)	Evening (+ R)
Mon	Z2H makemore (bigram → MLP LM)	T: NeetCode arrays/two-pointer (45m) → makemore	Re-watch; Anki; R: choose month-1 paper to reproduce
Tue	Z2H: BatchNorm internals	M: implement BatchNorm (45m) → makemore	Anki
Wed	Z2H: manual backprop ("backprop ninja")	T: binary search / sliding window → build	Anki
Thu	Attention: scaled dot-product + $\sqrt{d_k}$	M: implement scaled-dot-product attention	Anki (attention)
Fri	P PHASE-1 SELF-TEST (derive backprop; micrograd from memory; BatchNorm why; attention+ $\sqrt{d_k}$)	M: multi-head attention	R: start reproduce-paper
Sat-Sun	Buffer + rest (if self-test not clean → shore up here, shift a few days)		

Week 11 · Nov 16-22 — Z2H finish (GPT) + imaging kickoff · threads running

Day	Block A	Block B (T/M + build)	Evening (+ R)
Mon	Z2H: build a small GPT (blocks, pos-enc)	T: stacks/queues → GPT build	Imaging physics (MRI Q&A); R: reproduce-paper
Tue	GPT: attention blocks	M: sampler (top-k/temperature) → GPT	Anki
Wed	GPT: training loop	T: linked lists → finish GPT	Fluorescence primer (iBiology)
Thu	Tokenizer (BPE)	M: mini-BPE → tokenizer	Anki; R: write-up progress
Fri	Review the NN stack	T: hashing / two-sum patterns	Whiteboard-Fri: GPT data flow; note

Sat-Sun	Buffer + rest		
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Week 12 · Nov 23-29 — CONSOLIDATION + light threads

Day	Block A	Block B (+ light T)	Evening (+ R)
Mon-Fri	Re-derive (blank page): backprop, attention+ $\sqrt{d_k}$, softmax-CE gradient, SVD	Re-implement micrograd-MLP + multi-head attention; T : spaced review of solved problems	Anki catch-up; R : finish month-1 reproduce-paper + write-up; Fri: teach-a-junior note on attention
Sat-Sun	Rest + buffer		

End of Phase 1 → Phase 2 (Biomedical Imaging) next. Gate to advance: the Week-10 self-test is clean **and** you can, cold — derive SVD from eigendecomposition, hand-derive backprop, implement micrograd from memory, and whiteboard attention with $\sqrt{d_k}$.